

Design paradigms

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graph TD; A[Design paradigms] --> B[Divide & Conquer<br/>merge sort, quick sort,<br/>binary search,<br/>Strassen, closest pair,<br/>Karatsuba]; A --> C[Greedy<br/>Kruskal, Prim, Dijkstra,<br/>Huffman,<br/>fractional knapsack,<br/>activity selection, job<br/>sequencing]; A --> D[Dynamic Programming<br/>0/1 knapsack, LCS,<br/>matrix chain, Floyd-Warshall,<br/>Bellman-Ford, OBST,<br/>TSP Held-Karp]; A --> E[Backtracking<br/>N-Queens, sum of<br/>subsets, graph<br/>colouring,<br/>Hamiltonian cycle,<br/>Sudoku]; A --> F[Branch & Bound<br/>0/1 knapsack, TSP, job<br/>assignment - uses<br/>bounds, BFS/LC search];
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